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Lesson 6: k-NN, Naive Bayes and Support Vector Machines

Technologies for Artificial Intelligence

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Before we start

- Exercises 4 and 5, which we discuss now
- The threshold you chose; the score that was too good, and what the honest number cost you

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Today: the first lesson that offers a choice

- **k-nearest neighbours (k-NN)**, and the **curse of dimensionality** that limits it
- **Naive Bayes**: one strong assumption, a cheap model
- **Support vector machines (SVM)**: margins and kernels
- Which family to choose, and why that beats tuning

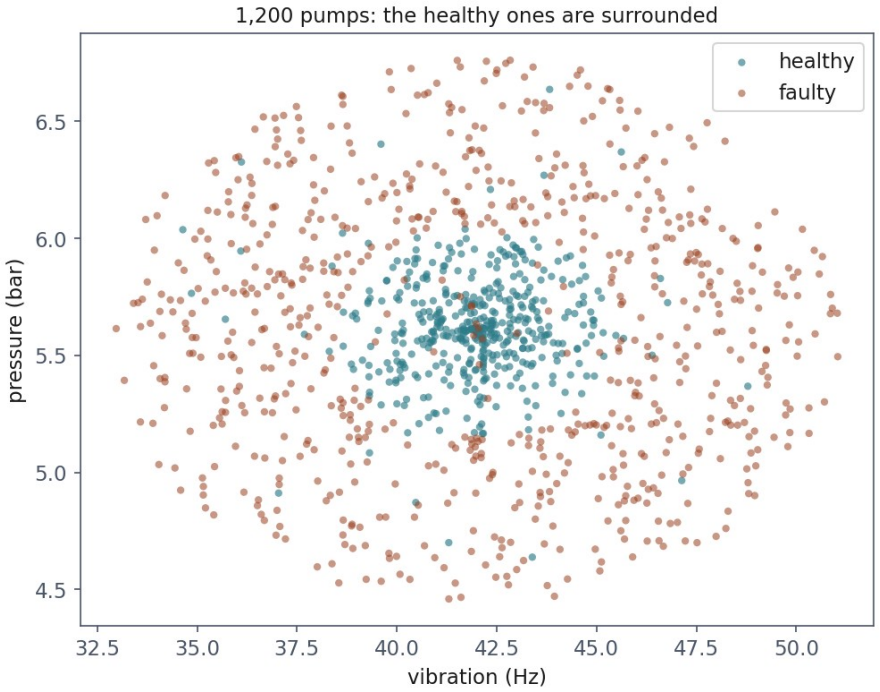
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1,200 pumps, two readings each

- Vibration in Hz, pressure in bar
- A pump is faulty when its readings fall **outside the design envelope** (the range it was built to run in), too low as readily as too high
- **61.3%** faulty; 4% of the labels are deliberately flipped

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The shape of the problem



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What a straight boundary costs

Model	Cross-validated accuracy
Always predict the majority class	0.613
Logistic regression (lesson 4)	**0.613 ± 0.000**

k-nearest neighbours: the whole algorithm

- To classify a new point, find the **k closest** training points and take the majority vote
- There is no training step: “fitting” means **storing the data**
- Called a **lazy learner**: an unusually honest name for an algorithm
- In code:

```
KNeighborsClassifier(n_neighbors=5)
```

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Distance is all it has

$$d(x, x') = \sqrt{\sum_{j=1}^{n} (x_j - x'_j)^2}$$

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Scale first, or vibration decides everything

- Vibration runs over **tens of Hz**; pressure over **a couple of bar**
- Unscaled, a one-bar pressure difference is invisible beside a ten-Hz one
- Lesson 2’s scaling argument, arriving with **immediate consequences**
- In code: `make_pipeline(StandardScaler(), KNeighborsClassifier(5))`

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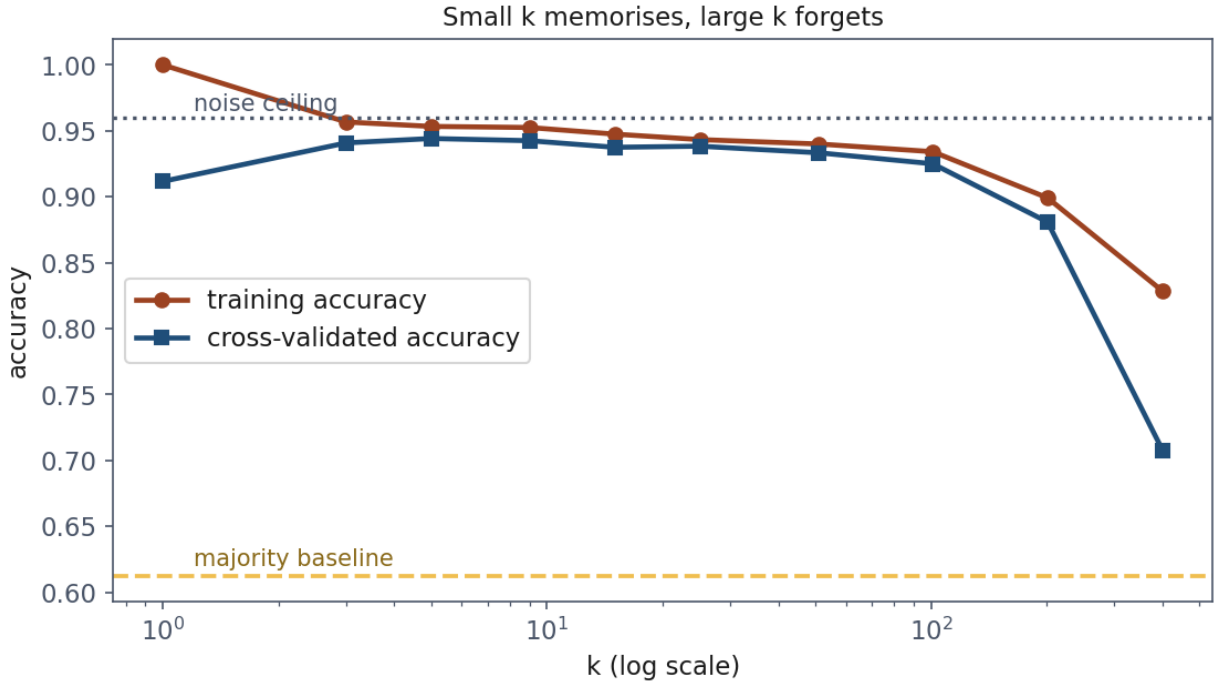
k is the bias-variance dial, made visible

- **Small k**: the boundary follows every point, mislabelled ones included. Low bias, high variance
- **Large k**: the vote is taken over a wide neighbourhood, and eventually stops following real structure. High bias, low variance
- At k = 1 the training accuracy is **exactly 1.000**, unless two identical points carry different labels

Measured on the pumps

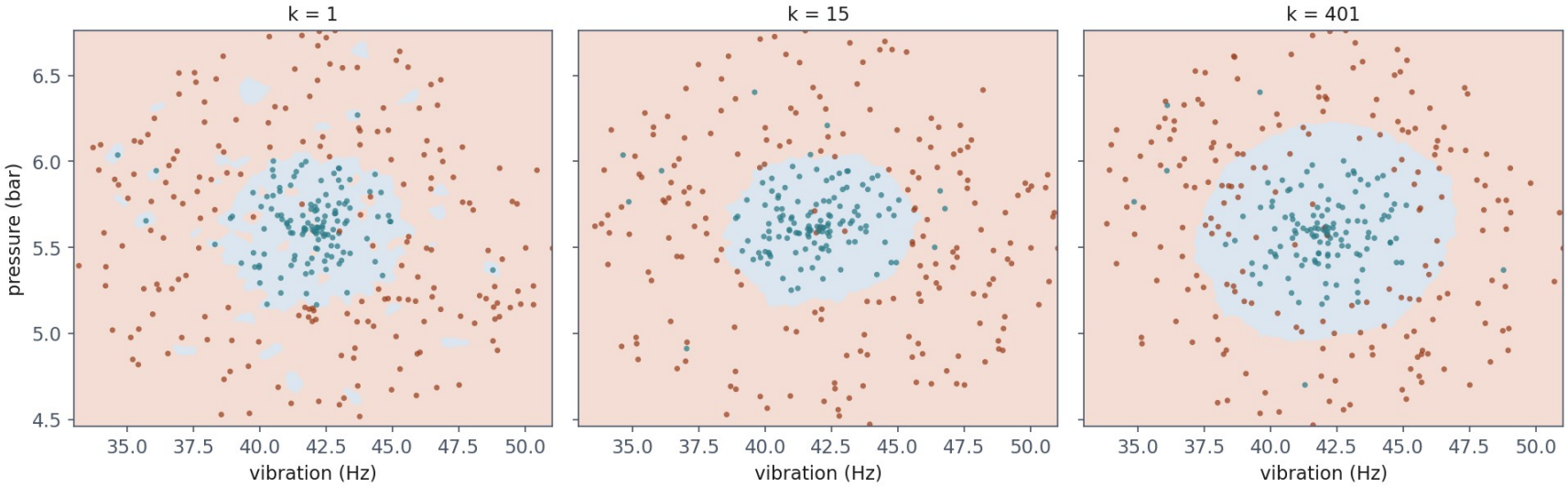
k	Training	Cross-validated
1	**1.000**	0.912
5	0.953	**0.944**
15	0.948	0.938
51	0.940	0.933
401	0.828	0.708

The two curves, and the two lines that bound them



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The same data at three values of k



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What k-NN costs

- No training time at all: you pay at **every prediction** instead
- A naive implementation compares the query against **every** training row: $O(mn)$ for m rows and n features
- **The model is the dataset.** You cannot ship one without the other

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Now the price: add columns of nothing

- Extra columns of **pure noise**, drawn from a normal distribution
- The two real readings are **untouched**: the problem is exactly as solvable
- Only the number of columns changes

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The signal never left

Total columns	Accuracy	Above baseline
2	0.938	+0.325
12	0.762	+0.149
27	0.662	+0.049
52	0.602	**−0.011**
102	0.578	−0.035

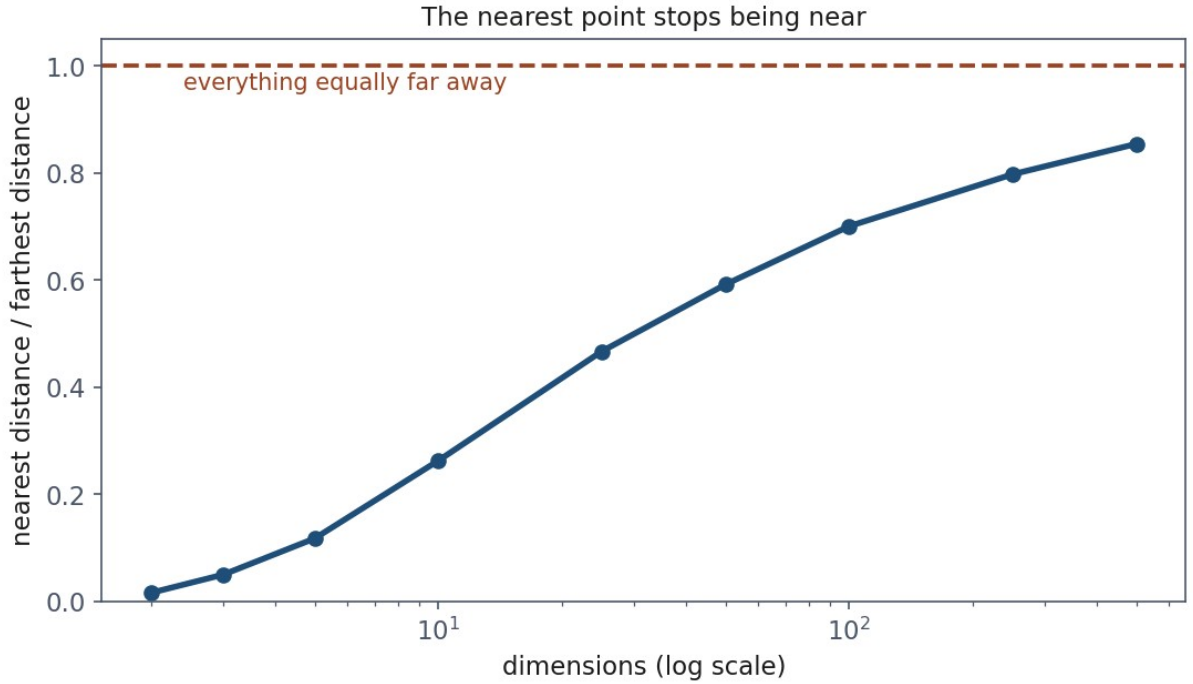
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Why: distances stop varying

- Scatter points in a cube, pick one, measure its **nearest** and its **farthest** neighbour
- In two dimensions those are very different numbers, which is what makes “nearest” a meaningful word
- Each new dimension adds its own squared difference, and the contributions **average out**

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The nearest point stops being near



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Nearest divided by farthest

Dimensions	nearest ÷ farthest
2	0.016
10	0.263
50	0.592
100	**0.701**
500	0.855

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In 100 dimensions the nearest point is 70% as far as the farthest

- In **two** dimensions it is **2%**: “nearest” picks out something special
- In **one hundred** it is **70%**: the nearest point is barely nearer than a random one
- A vote among “the five nearest” becomes a vote among five taken at random

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What the curse is, and is not

- **Not** that high-dimensional problems are unlearnable: lesson 9's networks work in thousands of dimensions
- That **methods built on distance lose their footing**, because the quantity they depend on stops varying
- k-NN, k-means (lesson 8), and the radial basis function (RBF) kernel of support vector machines, later today

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Notebook 1, live

section, and its slides	you call (→ the number to come back with)
the straight line, 4-6	`LogisticRegression` scored by `cross_val_score` over `StratifiedKFold` folds → 0.613
k-NN, scaled first, 7-9	`make_pipeline(StandardScaler(), KNeighborsClassifier(k))`
choosing k, 10-13	`cross_val_score` for each k → 0.944 at k = 5
columns of noise, 15-16	the same pipeline on `load_with_noise(d)` → 0.602 at 52 columns
distances stop varying, 17-20	NumPy only: nearest ÷ farthest → 0.701 in 100 dimensions

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Break

- Twelve minutes

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Naive Bayes: turn the question around

- We want the probability of the class **given** the readings
- The prior P(y = c) is the class's share of the data; the denominator is the same for every class and cancels
- Everything hard sits in one term: the probability of **this exact combination of readings** among examples of that class

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Bayes' rule, applied to a class

$$P(y = c \mid x) = \frac{P(x \mid y = c)\ P(y = c)}{P(x)}$$

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The decision drops the denominator

$$\hat{y} = \arg\max_c P(x \mid y = c)\, P(y = c)$$

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The assumption: independent, given the class

- **Given the class, the features are independent of one another**
- The joint density then factorises into n one-dimensional densities
- Each factor is estimated from the rows of that class alone
- In code: `GaussianNB()`

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The factorisation the assumption buys

$$P(x \mid y = c) = \prod_{j=1}^{n} P(x_j \mid y = c)$$

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Gaussian: each factor is a bell curve

$$P(x_j \mid y = c) = \frac{1}{\sqrt{2\pi\sigma_{jc}^2}} \exp\left[-\frac{(x_j - \mu_{jc})^2}{2\sigma_{jc}^2}\right]$$

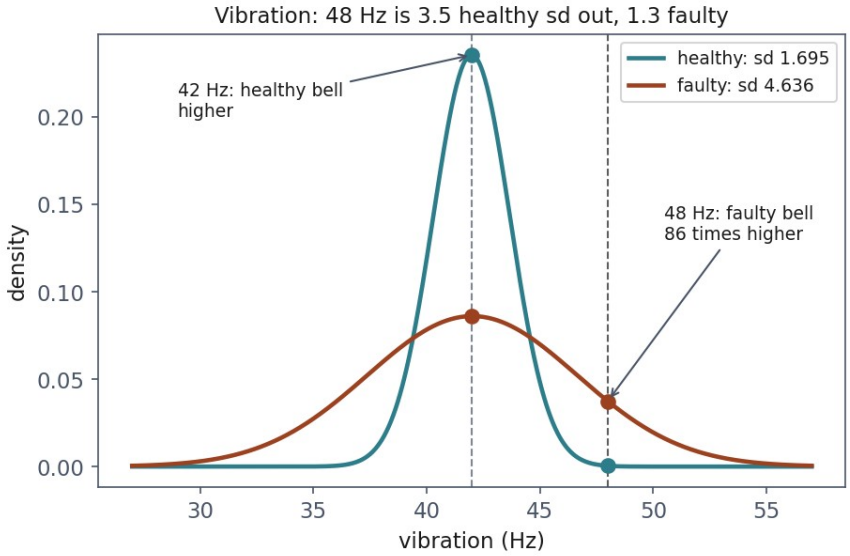
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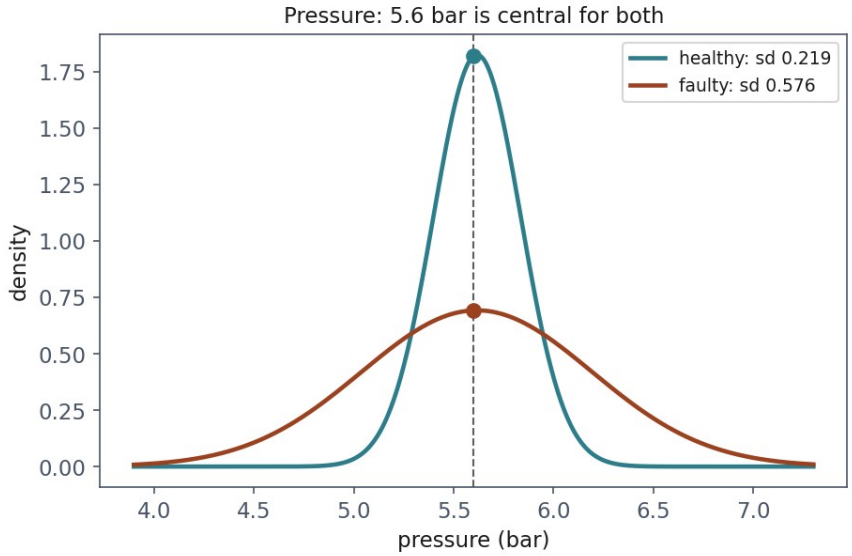
Ten numbers, and the widths decide

Class	Prior	Vibration mean	Vibration sd	Pressure mean	Pressure sd
Healthy	0.3875	41.986 Hz	**1.695**	5.617 bar	**0.219**
Faulty	0.6125	42.015 Hz	**4.636**	5.616 bar	**0.576**

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48 Hz: one bell is 86 times the other





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Add the natural logs: same winner

$$\ln\left[P(y=c)\prod_j P(x_j \mid y=c)\right] = \ln P(y=c) + \sum_j \ln P(x_j \mid y=c)$$

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48 Hz, 5.6 bar: scores and their logs

Class		Prior	Vibration density	Pressure density	Product, or sum of logs
Healthy	value	0.3875	0.000433	1.818	0.000305
Healthy	ln	−0.948	−7.745	+0.598	−8.095
Faulty	value	0.6125	0.0374	0.692	0.01585
Faulty	ln	−0.490	−3.286	−0.368	**−4.144**

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From the two totals to probabilities

Step	Healthy	Faulty
1. Log total, previous slide	−8.095	−4.144
2. Undo the log: e^total = prior × densities	0.000305	0.01585
3. Divide by their sum: 0.01616 = P(x), by total probability	**0.019**	**0.981**

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Why the gap is enough

- Divide by the faulty score: 1 / (1 + healthy score ÷ faulty score)
- e^a ÷ e^b = e^(a − b), so the ratio is e^(−3.951) = 0.0192
- P(faulty) = 1 / 1.0192 = **0.981**; P(healthy) = 1 − 0.981 = **0.019**
- Odds: 0.01585 ÷ 0.000305 ≈ **52 to 1**
- With many features e^total rounds to 0 in the computer; the gap does not

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A wonderful bargain, and almost never true

- Training is **a single pass**; adding features costs almost nothing
- But vibration and pressure are both driven by the operating point
- “New” is not independent of “York” given the topic. Symptoms co-occur
- So the question is not whether it holds: it is **when being wrong about it costs you nothing**

On the pumps, it scores 0.933

Model	Accuracy
Majority baseline	0.613
Logistic regression	0.613
Gaussian Naive Bayes	**0.933**
k-NN, k = 5	0.944

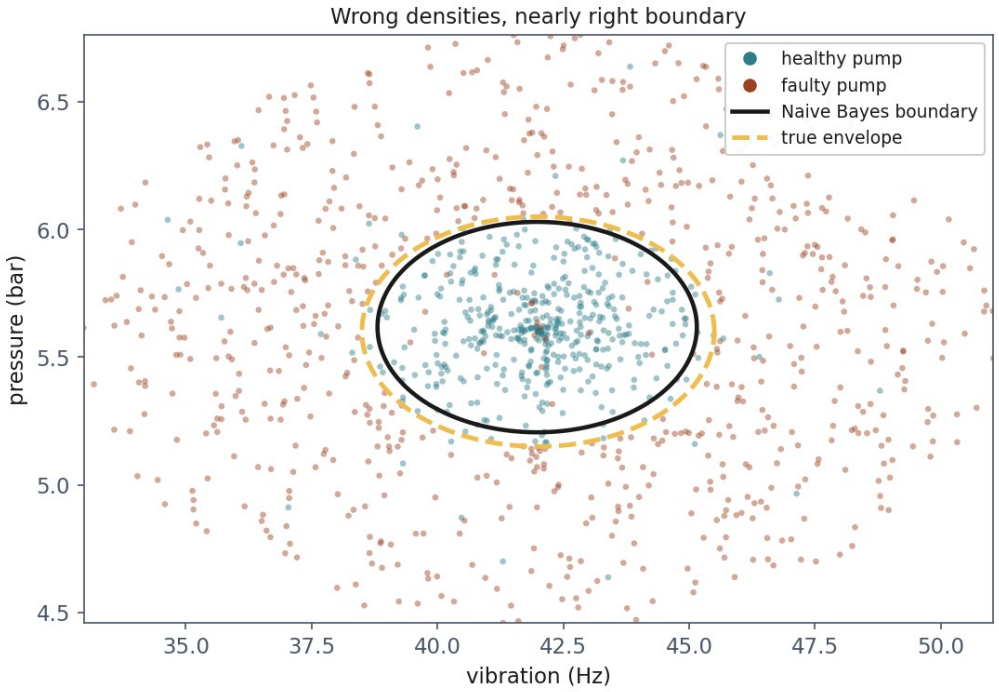
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Uncorrelated, but not independent

Correlation, within the class	vibration and pressure	their squared distances from the design point
healthy pumps	−0.006	+0.168
faulty pumps	−0.049	**−0.420**

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Wrong densities, right boundary

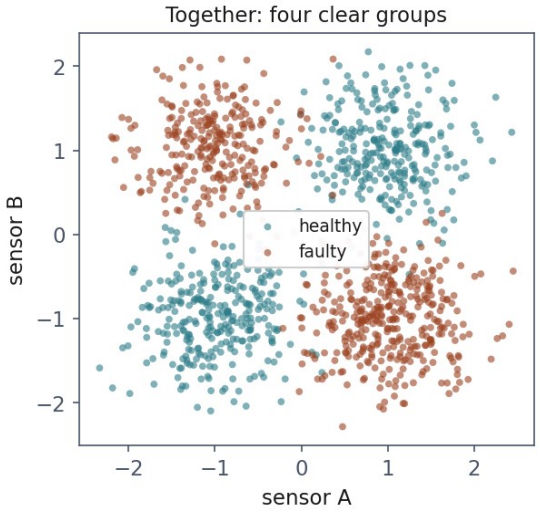


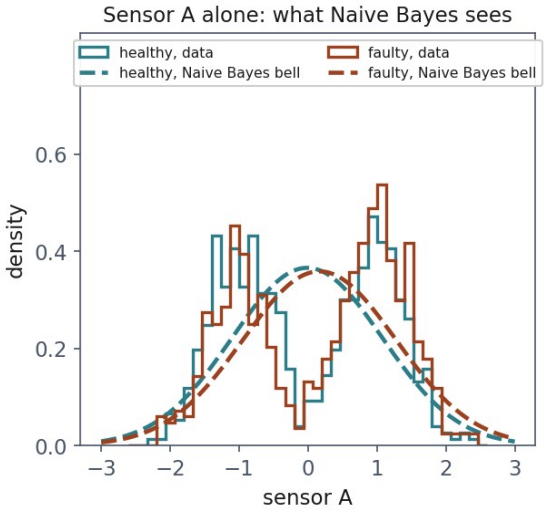
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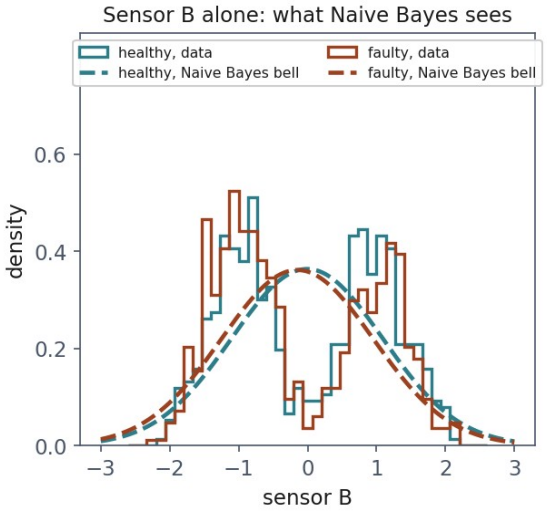
One step away: when the signal is an interaction

- A second pair of sensors on the same fleet
- Faulty when **exactly one** of the two readings is high
- Both high is the designed high-load mode; both low is idle; one without the other is a mismatch between demand and delivery

What Naive Bayes gets to see







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0.404: below the baseline, and below chance

Model	Accuracy
Majority baseline	0.523
Gaussian Naive Bayes	**0.404**
Logistic regression	0.393
SVM, linear kernel	0.606
k-NN, k = 5	0.967
SVM, RBF kernel	0.972

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Its probabilities are not probabilities

- Confidence, the probability given to the predicted class: **0.567** when correct, **0.555** when wrong
- It cannot tell the difference between the two situations
- With features that repeat each other it fails the other way: 0.999 reported, with an accuracy nothing like that
- **The ranking may be useful while the probabilities are not**

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Notebook 2, live

section, and its slides	you call (→ the number to come back with)
on the pumps, 37	`GaussianNB` beside `KNeighborsClassifier` and `LogisticRegression` → 0.933
what training computed, 29-35	`.theta_`, `.var_`, `.class_prior_`, `predict_proba` → 0.981 at 48 Hz
checking the assumption, 38-39	`X[y == c].corr()`, within each class → −0.420
the interacting sensors, 40-42	the same models, plus `SVC` → 0.404
confidence, right and wrong, 43	`train_test_split`, then `predict_proba` → 0.567 against 0.555

The margin: a criterion that is not the error

- On separable data there are **infinitely many** separating lines, and every one has **zero training error**
- Minimising the error therefore cannot choose between them
- The SVM picks the widest **slab**: push it out until it touches the nearest point of each class
- A boundary passing close to a point is one small perturbation from getting it wrong

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Three points decide everything



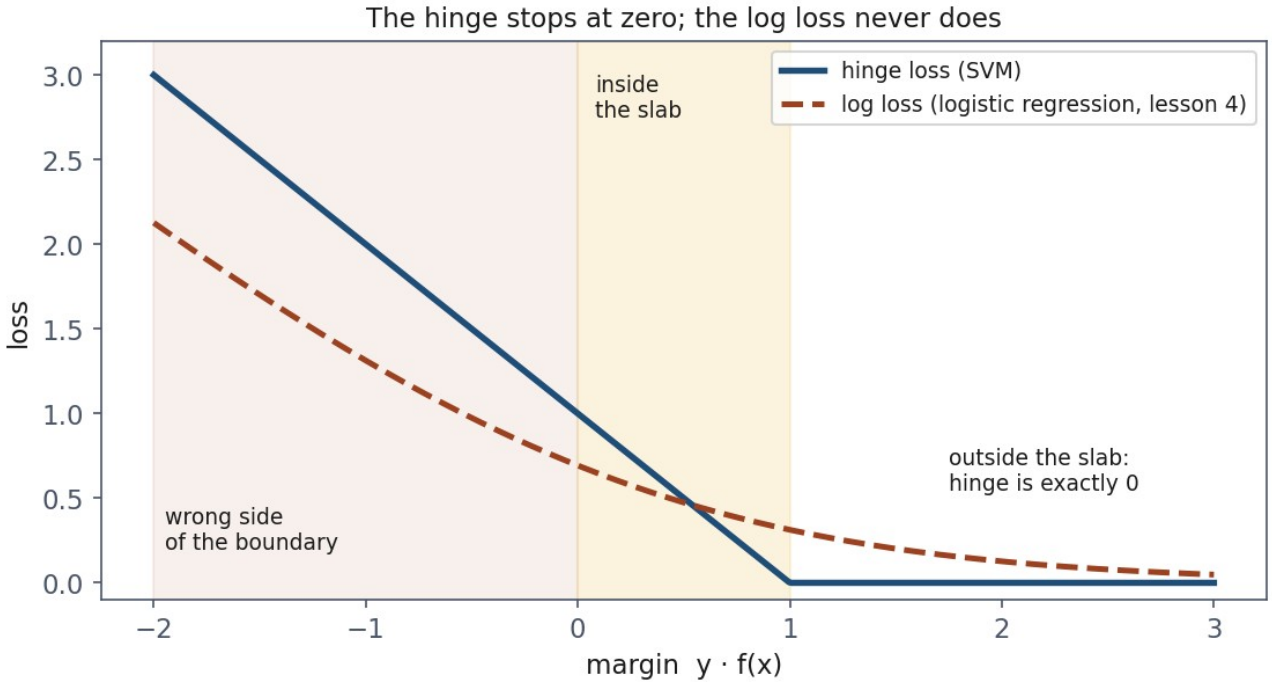
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Not separable? Charge for the violations

- Our labels are 4% flipped, so the strict problem has **no solution at all**
- Let each point violate the margin by ξ_i, and charge C for every unit of it
- $\xi_i = 0$ outside the slab, between 0 and 1 inside it, above 1 on the wrong side
- That is the **hinge loss**, $\max(0, 1 - y \cdot f(x))$ with labels ± 1: zero for a point with room to spare

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The hinge stops at zero



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A wide slab, minus what the violations cost

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_i \xi_i$$

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C is the price of a training error

- **Large C**: violations are expensive, so the model accepts a narrower margin to classify more training points. Low bias, high variance
- **Small C**: a wider, calmer boundary, at the cost of some errors
- C plays the role that k played in k-NN and λ in lesson 3
- Note the direction: **large C means less regularisation**

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The best straight line is still a straight line

Model	Accuracy	Support vectors
SVM, linear kernel	**0.613 ± 0.000**	947 of 1,200 (**79%**)
SVM, RBF kernel	0.947 ± 0.005	278 of 1,200 (23%)

A high support-vector fraction is a free warning

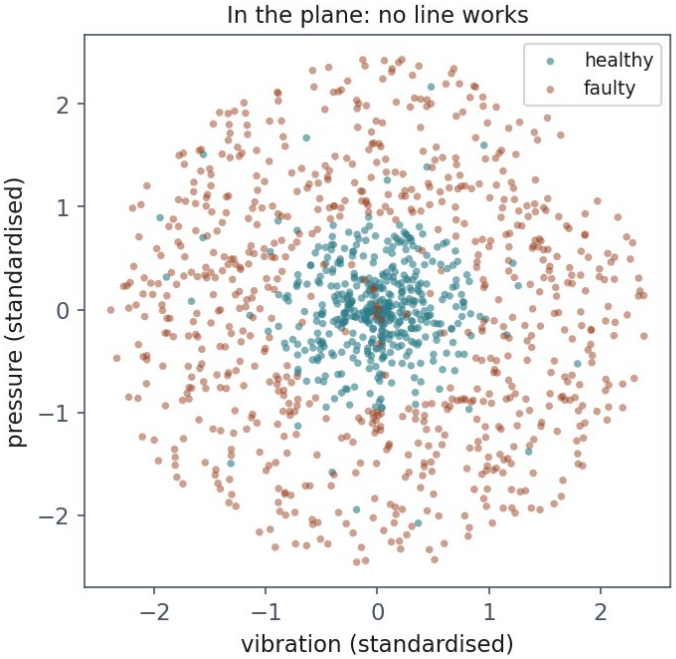
- The linear model, `SVC(kernel="linear")`, needs **79%** of the training set
- The RBF model needs **23%**
- Almost every point sits on or inside the margin: there is no slab that separates anything
- The count comes free with `fit`, as `len(model.support_)`: look at it

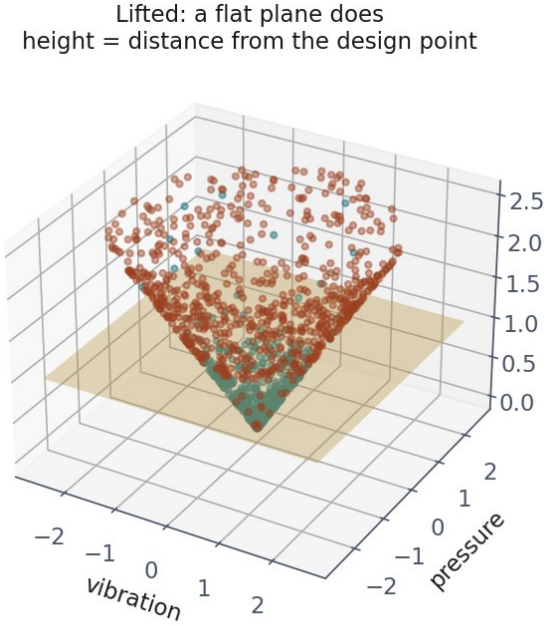
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The kernel trick: they needed different coordinates

- Healthy pumps in a disc, faulty ones around it: no line works **in the plane**
- Add a third coordinate: the **distance from the design point**
- Healthy pumps rise a little, faulty ones a lot, and a flat plane separates them perfectly
- The classes were always separable. They were measured in the wrong coordinates

The same pumps, lifted





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And the map is never computed

- The solution depends on the data **only through inner products** between pairs of points
- So we need a function returning the inner product **in the new space**, while computing only with the original one
- The radial basis function is scikit-learn’s default: one exponential per pair, for an **infinite-dimensional** space

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The boundary is a weighted vote of the pumps

$$w = \sum_{i=1}^{m} a_i \, y_i \, x_i, \qquad a_i \geq 0$$

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The kernel trick by hand: 25 = 25

Route	Computation	Result
Lift, then multiply	$(1, 2.83, 4) \cdot (9, 4.24, 1) = 9 + 12 + 4$	**25**
Never lift	$((1, 2) \cdot (3, 1))^2 = 5^2$	**25**

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The radial basis function, written out

$$K(x, x') = \exp\left(-\gamma \|x - x'\|^2\right)$$

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Training: one weight per pump

- Weights adjusted **two at a time**; no learning rate, same answer every run
- **922** pumps: weight 0, forgotten
- **268** at the ceiling C = 1, **10** on the edge: the 278 support vectors
- The model: 278 points, their weights, and b = +1.105

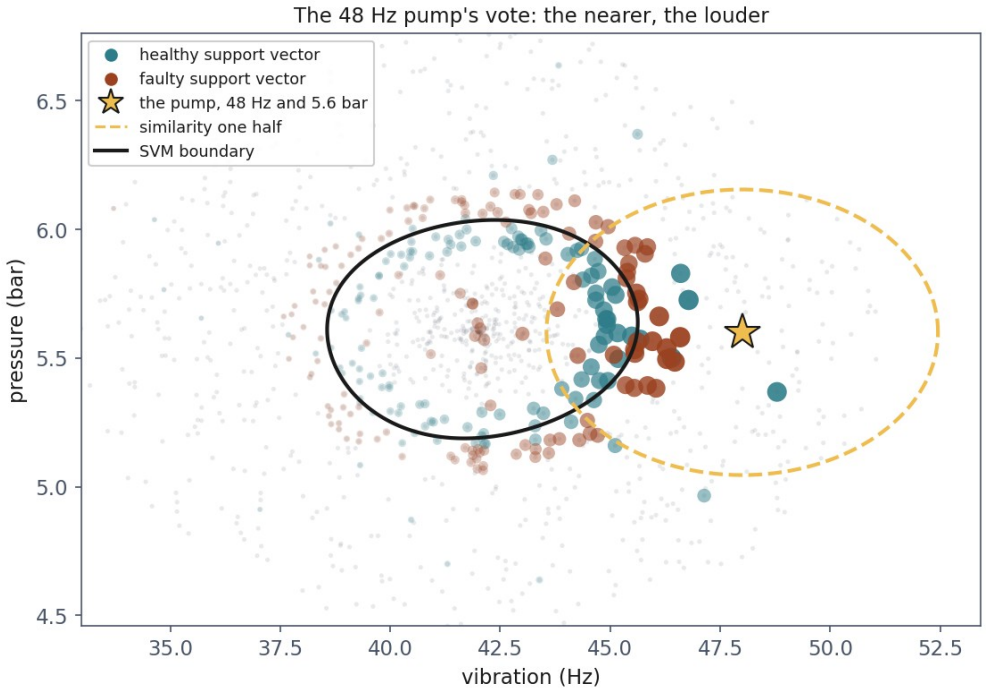
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The SVM on 48 Hz, 5.6 bar: +1.883

Term	Value
votes from faulty support vectors	+37.455
votes from healthy support vectors	−36.677
intercept b	+1.105
total	**+1.883**

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One pump’s vote, drawn

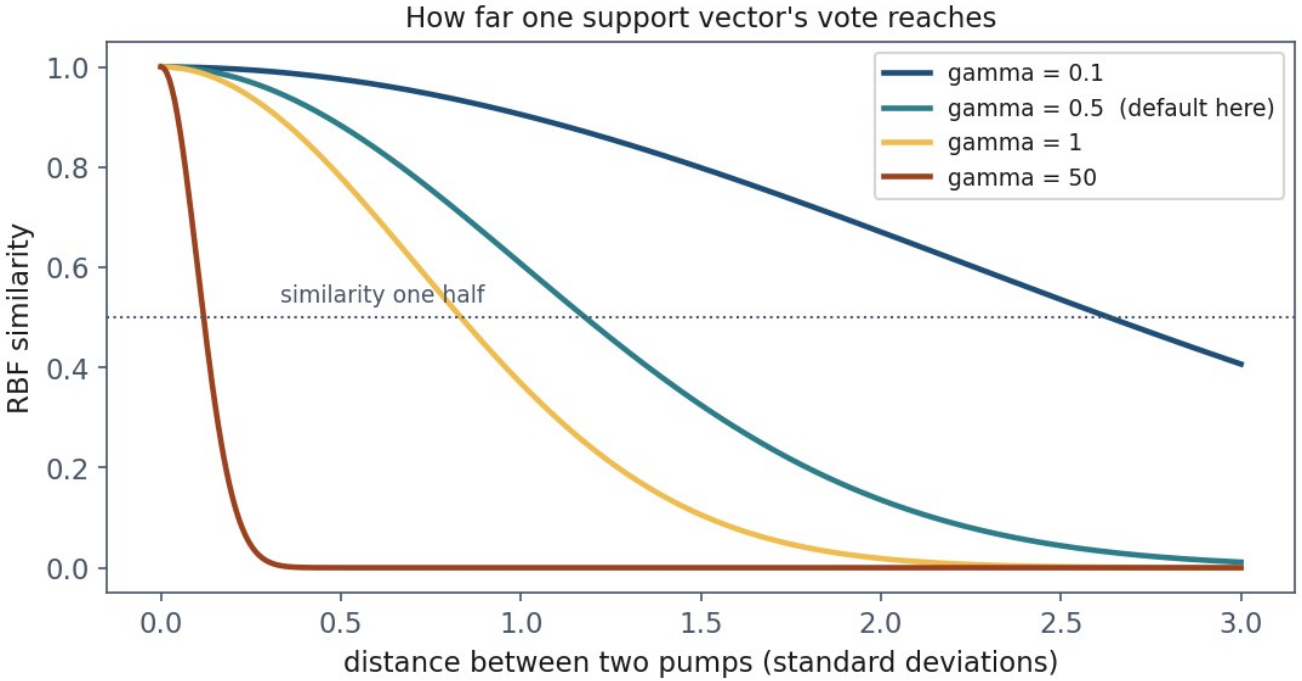


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The highest training score, the lowest honest one

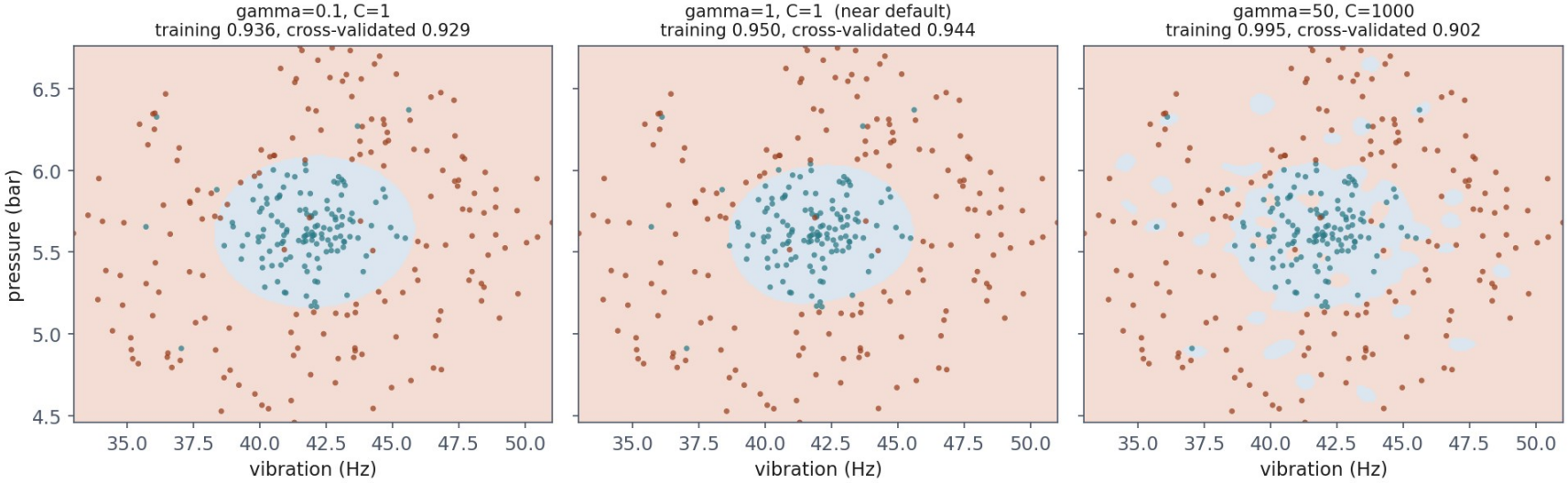
Setting	Training	Cross-validated
$\gamma = 0.1, C = 1$	0.936	0.929
$\gamma = 1, C = 1$	0.950	**0.944**
$\gamma = 50, C = 1000$	**0.995**	0.902

How far one vote reaches



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Overfitting you can see



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Notebook 3, live

section, and its slides	you call (→ the number to come back with)
the widest slab, 45-48	`SVC(kernel="linear")`, `.n_support_`, `decision_function` → 3 of 80 points
a straight line still fails, 51-52	`cross_val_score`, linear against RBF kernel → 947 against 278 support vectors
the kernel trick, 53-58	NumPy only: $(a \cdot b)^2$ against the lifted product → 25 = 25
training and one prediction, 59-61	`.support_`, `.dual_coef_`, `.intercept_` → +1.883
gamma and C, 62-64	`SVC(gamma=..., C=...)` in the pipeline → 0.944 at $\gamma = 1$

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The three compared, on identical data

Model	Accuracy
Majority baseline	0.613
Logistic regression (lesson 4)	0.613
SVM, linear kernel	0.613
Gaussian Naive Bayes	0.933
k-NN, k = 5	0.944
SVM, RBF kernel	**0.947**
noise ceiling	*≈ 0.96*

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How to choose, in practice

Situation	Reach for
Few features, plenty of data, odd-shaped boundary	k-NN, or an RBF SVM
Very many features, little data	Naive Bayes
Need a calibrated probability	**Not** Naive Bayes
Model must be small, or fast at prediction	SVM, not k-NN
The signal is an interaction between features	k-NN or a kernel

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What to take away

- **No line separates the pumps**: 0.613, the base rate
- **k is the bias-variance dial**, made visible
- **In 100 dimensions the nearest point is 70% as far as the farthest**
- **Naive Bayes**: 0.933 where its false assumption is cheap, **0.404** where not
- **Choosing the family beats tuning the wrong one**: 0.613 to 0.947

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Homework

- **Exercise 6**, discussed at the start of **Friday 20 November**
- A dataset, three families, and a defence of which one you chose

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